Fentanyl

Mechanism of action

- Fentanyl is an opiate analgesic. It is an opiate agonist (or stimulator) that binds to opiate receptors in the brain and spinal cord causing analgesia.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ Moderate to severe pain.
- ✓ Cardiogenic pulmonary oedema with severe anxiety.
- ✓ Rapid sequence intubation.
- ✓ Sedation post intubation.
- ✓ Control of pain, agitation, or shortness of breath during the last days of life.
- ✓ Autonomic dysreflexia, even if the patient cannot feel pain.

Contraindications

- ✗ Known severe allergy.

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Cautions

- Altered level of consciousness. Fentanyl may further reduce the level of consciousness.
- Aged less than one year. Children under the age of one year are at increased risk of respiratory depression following opiate administration.
- Respiratory depression or at high risk of respiratory depression. For example, severe COPD, morbid obesity or on home BiPAP. Such patients may develop respiratory depression following opiate administration.
- Labour. Opiates cross the placenta and may cause drowsiness and/or respiratory depression in the baby, particularly when administered within an hour or two of birth. Discuss administration with the lead maternity carer if possible. Following birth, close observation of the baby is required and personnel must be prepared to treat respiratory depression.
- Concurrent administration of other opiates, ketamine or midazolam. This will increase and prolong the effects.
- Aged greater than or equal to 75 years, particularly if frail. The effects of fentanyl will be increased and prolonged in this cohort.
- Signs of shock. Fentanyl may make shock worse.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated in pregnancy, but fentanyl should be administered if indicated.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- IV bolus for analgesia:
  - 10-50 mcg every five minutes for an adult, as required. Use a dose at the lower end of the range if the patient is frail or has signs of shock.
  - See the paediatric drug dose tables for a child.
- IV infusion:
  - Start at 1 mcg/kg/hour, up to a maximum of 150 mcg/hour.
  - Administer additional boluses as required, every five minutes.

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- Adjust the infusion rate as required, every 15-20 minutes.

- IV for autonomic dysreflexia:
  - 20-30 mcg increments to a total of 100 mcg.

- IN for analgesia:
  - 100 mcg for an adult.
  - See the paediatric drug dose tables for a child.
  - Halve the dose if the patient is frail or has signs of shock.
  - Repeat as required every 20 minutes.

- IM and SC for analgesia:
  - 50-100 mcg IM/SC for an adult.
  - See the paediatric drug dose tables for a child.
  - Halve the dose if the patient is frail or has signs of shock.
  - Repeat as required, every 20 minutes, up to a maximum of three doses.

- IV for RSI: see the **'RSI'** guideline.

- IV for sedation post intubation: see the **'post intubation'** guideline.

Administration

- The preferred route for administration is IV.

- IV boluses for analgesia:
  - Dilute 100 mcg to a total of 10 ml using 0.9% sodium chloride for an adult or a child whose weight has been rounded up to 30 kg or more. This final solution contains 10 mcg/ml.
  - Dilute 100 mcg to a total of 100 ml using a 100 ml bag of 5% glucose for a child whose weight has been rounded up to 20 kg or less. This final solution contains 1 mcg/ml.

- IN administration:
  - Draw up fentanyl undiluted, placing half of the total dose into two separate 1 ml syringes. When drawing up the first syringe, draw up an additional 0.1 ml of fentanyl over and the above the planned volume and expel this slowly through the mucosal atomiser in order to fill the dead space. This does not need to be done with subsequent doses.
  - Administer fentanyl IN by rapidly injecting one syringe (half of the total dose) into each nostril. Rapid injection is required to achieve a fine mist, which maximises absorption.

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- IM administration: Administer undiluted. The preferred IM site is the lateral thigh. If this site is not suitable use the lateral upper arm.
- SC administration:
  - Administer undiluted subcutaneously into the abdominal wall.
  - There is no need to sterilise the skin at the site of injection unless the skin is visibly contaminated.
  - Pinch a fold of skin over the anterior abdominal wall between thumb and forefinger. Introduce the entire length of the needle using a dart technique and inject.

Common adverse effects

- Respiratory depression.
- Bradycardia.
- Hypotension.
- Sedation.
- Nausea and vomiting.
- Itch.
- Euphoria.

Usual onset of effect

- IV: 2-5 minutes. The maximal analgesic and respiratory depressant effects may not occur until 10-15 minutes and this may be longer in the elderly.
- IN: 5-10 minutes.
- IM/SC: 5-10 minutes.

Usual duration of effect

- 30-60 minutes.
- The effect on respiration may last for several hours.

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Usual preparation

- Ampoule containing 100 mcg in 2 ml.

Pharmacokinetics

- Fentanyl is more lipophilic (fat soluble) than morphine and this is why fentanyl is well absorbed through the nasal mucosa.
- Fentanyl may cause a small amount of histamine release. In combination with relief of pain this usually results in a small fall in blood pressure.
- Fentanyl is metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- The effects will be increased in the presence of other opiates and sedatives, for example, benzodiazepines or alcohol.

Additional information

- A patient administered fentanyl must be given a clear recommendation to be transported to a medical facility by ambulance, unless the patient is being treated using the 'care during the last days of life', 'patella dislocation' or 'shoulder dislocation' guidelines.
- Transport should usually be to an ED unless it is appropriate for the patient to be transported to a primary care facility.
- Fentanyl is usually the preferred opiate if:
  - The patient requires analgesia for a short period of time, or
  - The patient has renal impairment, or
  - The patient has signs of shock.

More Information

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Version history

16 June 2026 1.1.0.1

Update: Dosage updated.

22 November 2022 1.0.1

Update: Contraindications, cautions and aligned with analgesia and end of life CPGs.